

555 Clock Module

Clock Pulse Generator · NE555P Astable Oscillator

REVISION

A

DESIGNED BY

Mohammad Aljumaah

DATE

2026-06-24

STATUS

Released

1 Overview

The **555 Clock Module** is a small board that produces a steady blinking signal — like a heartbeat — which other circuits can use to keep time or stay in sync. It is built around the classic **NE555P** chip, which has been used in electronics projects for decades.

The board is intentionally simple: three components (two resistors and one capacitor) set the speed of the signal, and a 3-pin connector lets you plug it straight into a breadboard or another circuit. All parts are standard through-hole components, so they are easy to solder by hand.

2 Specifications

Parameter	Value	Notes
Power Required	5 V DC	Via J1 Pin 2 (VCC)
Output Signal	Square Wave (on/off pulses)	Astable mode
Output High Voltage	~3.3 V	When signal is ON
Output Low Voltage	~0 V	When signal is OFF
Speed (Frequency)	~6–7 blinks per second	With included component values
On/Off Balance	~50% on, 50% off	Equal on and off time
Output Connector	3-Pin Header (J1)	GND / VCC / CLK_OUT
Main Chip	NE555P	8-pin DIP package
PCB Layers	2	Standard FR-4 board
Mounting Holes	4 × M3	One at each corner

3 How to Use

Setting up this module takes less than a minute. Follow these steps:

- 1 **Connect Ground (GND)**

Plug a wire from Pin 1 on J1 to the ground (negative) rail of your circuit or power supply.
- 2 **Connect Power (VCC)**

Plug a wire from Pin 2 on J1 to a 5 V power source. Do not exceed 5 V.
- 3 **Connect Your Device to CLK_OUT**

Plug a wire from Pin 3 on J1 to whatever you want the clock signal to control — a counter chip, an LED, a shift register, etc.

4 Power On

Switch on your power supply. The CLK_OUT pin will immediately start producing a steady on/off signal at roughly 7 times per second.

5 Check the Signal (Optional)

If you have a multimeter or oscilloscope, you can probe the CLK_OUT pin to confirm the signal is present.

■ Make sure your power supply is set to exactly 5 V before connecting anything.

✓ The CLK_OUT signal is ready the moment power is applied — there is no button or switch.

4 What You Can Connect This To

This module works as a clock source for many common beginner and intermediate projects. Here are some examples:

Counter Chip (e.g. 74HC4017)	Feed the clock signal in and watch LEDs light up one by one in sequence — great for a chaser light effect.
7-Segment Display Counter	Connect through a counter chip and watch a number go up automatically, one step per clock pulse.
LED Blinker	Connect an LED (with a resistor) to CLK_OUT and it will blink at roughly 7 times per second.
Shift Register (e.g. 74HC595)	Use this as the clock pin to shift data one bit at a time into a chain of outputs.
Oscilloscope or Logic Analyzer	A useful reference signal for learning how to read waveforms.
Microcontroller Input	Feed the signal into a digital input pin to test interrupt or timer-counting code.

5 Changing the Speed

The speed of the clock signal depends on the two resistors (R1 and R2) and the capacitor (C1). You do not need to understand the maths — just follow these simple rules:

What you want	What to change	Effect
Slower signal	Use a larger R2 (e.g. 47 kΩ instead of 10 kΩ)	Roughly 5× slower
Faster signal	Use a smaller R2 (e.g. 2.2 kΩ instead of 10 kΩ)	Roughly 5× faster
Much slower	Use a larger C1 (e.g. 100 μF instead of 10 μF)	Roughly 10× slower
Much faster	Use a smaller C1 (e.g. 1 μF instead of 10 μF)	Roughly 10× faster

What you want	What to change	Effect
Fine control	Replace R2 with a potentiometer (variable resistor)	Dial in the exact speed you want

■ Always keep R1 much smaller than R2. If R1 is too large, the on/off timing will become unequal.

6 Safety Notes

This module is safe to handle and use in normal conditions. Please follow these points to avoid damaging it or your other components:

■ **Do not exceed 5 V:** Supplying more than 5 V can permanently damage the NE555P chip and other connected components.

■ **Check the capacitor polarity:** The capacitor C1 has a positive (+) and negative (–) leg. Installing it the wrong way around can cause it to fail or burst.

■ **Do not short the output:** Avoid connecting CLK_OUT directly to GND or VCC. This will not damage the 555 instantly, but it is not good practice.

✓ **Handle with care:** Even though this is a simple board, avoid bending the pins or dropping it on hard surfaces.

✓ **Static sensitivity:** The NE555P is fairly robust, but try not to touch the chip pins directly. Ground yourself first if you can.

7 Known Limitations

This module is great for learning and prototyping. However, it does have a few things worth knowing about:

◆ The speed can drift slightly with temperature

The NE555P is not perfectly stable. If the room gets warmer or cooler, the signal speed may shift a tiny amount. For most projects this does not matter, but it is not suitable as a precision clock for things like real-time clocks or audio.

◆ It cannot drive heavy loads directly

The CLK_OUT pin can only power small things directly — like one LED or the input of a logic chip. Connecting it to something that draws a lot of current (like a motor) will cause it to misbehave or stop working.

◆ The frequency is fixed unless you swap components

There is no knob or button to change the speed. You need to physically swap R2 or C1 to change it (unless you add a potentiometer as described in Section 5).

◆ Not suitable for high-speed applications

With the included component values, the signal runs at about 7 blinks per second. The 555 can go much faster, but this board is designed for low-speed, easy-to-see-with-the-eye applications.

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Schematic

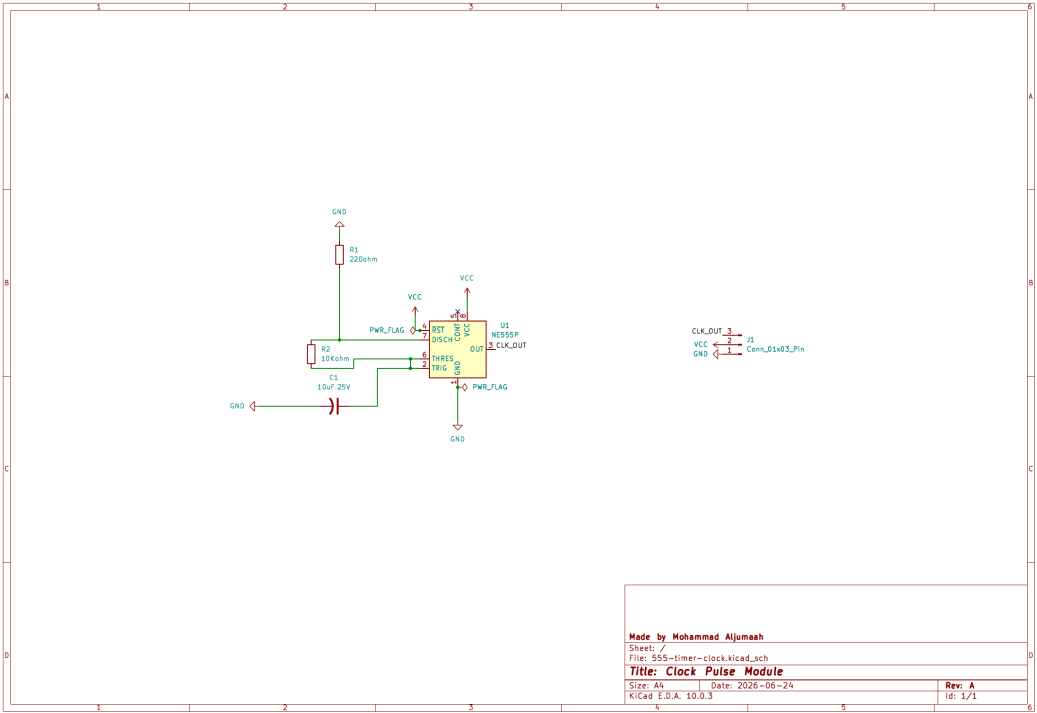


Figure 1 — Schematic (KiCad EDA 10.0.3)

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PCB Layout

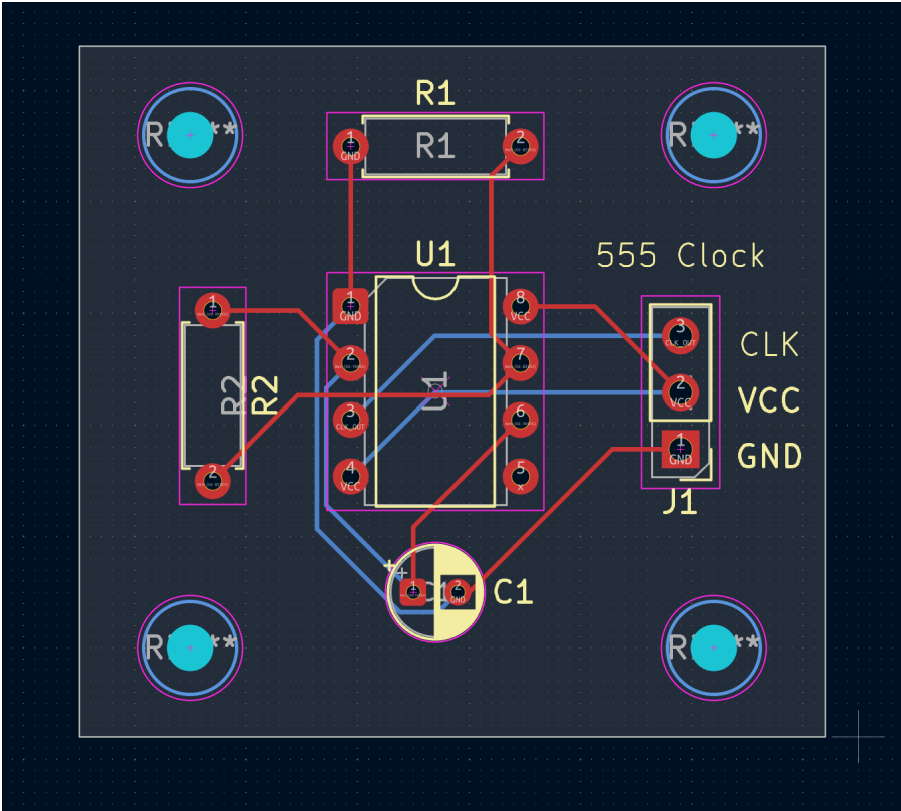


Figure 2 — PCB Layout (KiCad, 2-Layer View)

Item	Value
Layers	2 (Front copper / Back copper)
Surface Finish	HASL (default)
Silkscreen	Front & Back
Mounting Holes	4 × M3 (non-plated)
Design Tool	KiCad EDA 10.0.3
Revision	A

10 Bill of Materials

Ref	Component	Value	Package	Qty
U1	NE555P Timer Chip	NE555P	8-pin DIP	1
R1	Resistor	220 Ω	Axial	1
R2	Resistor	10 k Ω	Axial	1
C1	Electrolytic Capacitor	10 μ F 25 V	Radial ■5 mm	1
J1	Pin Header	1×3, 2.54 mm	Through-hole	1
MH1–MH4	Mounting Hole	M3 (non-plated)	—	4

11 Connector Pinout — J1

Pin	Name	Direction	Description
1	GND	Power	Ground — connect to negative of your power supply
2	VCC	Power In	5 V power input
3	CLK_OUT	Output	The clock signal comes out here

The connector uses a standard 2.54 mm spacing — the same as most breadboards and Dupont cables.

12 Glossary

Here are plain-English explanations for the words used in this document:

Astable

A mode where the circuit keeps switching on and off by itself, forever, without any outside trigger. Like a light that blinks automatically.

CLK_OUT

Short for "Clock Output" — the pin where the on/off signal comes out of this board.

Duty Cycle

The percentage of time the signal is ON versus OFF. A 50% duty cycle means it is on half the time and off the other half — perfectly balanced.

Frequency

How many times per second the signal switches on and off. This board runs at about 7 times per second, which is written as 7 Hz.

GND

Short for "Ground." This is the negative side of your power supply and the common reference point all voltages are measured from.

Hz (Hertz)

The unit for measuring frequency. 1 Hz means one complete on/off cycle per second. 7 Hz means seven cycles per second.

NE555P

The name of the main chip on this board. It is a classic, inexpensive timer chip that has been used in electronics since the 1970s.

Square Wave

A signal that switches sharply between fully ON and fully OFF — like a light switch, not a dimmer. This is the shape of the signal this board produces.

VCC

The positive power supply pin. On this board it needs to be 5 V.

13 How the Circuit Works

The NE555P chip is wired so that it charges and discharges the capacitor C1 back and forth through the resistors. Every time C1 fills up, the chip switches the output OFF. Every time C1 empties, the chip switches the output ON again. This repeating charge/discharge cycle produces the clock signal.

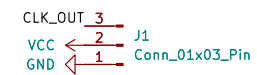
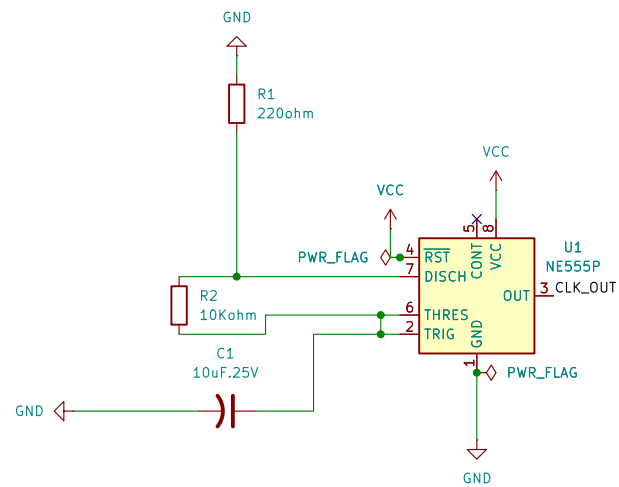
R1 and R2 control how fast C1 charges. R2 alone controls how fast it discharges. Because R2 is much larger than R1, both times are nearly equal — giving the 50/50 on/off balance.

Step	What Happens
C1 charges up	Current flows through R1 and R2 into C1. Output is ON.
C1 reaches the top threshold	The 555 detects this and switches the output OFF, then starts draining C1.
C1 drains down	Current flows back out of C1 through R2. Output stays OFF.
C1 reaches the bottom threshold	The 555 detects this and switches the output ON again, starting the next charge.
Cycle repeats	This loop runs continuously for as long as power is applied.

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Revision History

Rev	Date	Author	Description
A	2026-06-24	Mohammad Aljumaah	Initial release — through-hole astable 555 oscillator



Made by Mohammad Aljumaah

Sheet: /

File: 555-timer-clock.kicad_sch

Title: Clock Pulse Module

Size: A4

Date: 2026-06-24

Rev: A

KiCad E.D.A. 10.0.3

Id: 1/1